

Question Number	Answer	Additional guidance	Mark
2(a)	A calculation in which: - conversion of actual length of scale into μm (1) - calculation of magnification to two significant figures (1)	<u>Example of calculation:</u> 29 to 30 mm = 29 000 to 30 000 (μm) $(29\,000 \div 20) = \times 1450 = \times 1400$ OR 1500 incorrect unit = max 1 please note they may convert the 20 μm into mm or cm e.g. measurement $\div 0.02$ ecf for correct measurement incorrectly converted in working, $\div 20$ and then given to 2 sig fig, e.g. 150 ecf for everything correct with an incorrect measurement shown in working (remember 5 is rounded down OR up) Accept correct standard form to 2 sig fig Correct answer scores full marks	(2)

these are most common incorrect responses – but please don't rely on this – always use the MS as your guide

1 mark	2 marks
1450	1400 1.4×10^3
ecf for correct measurement incorrectly converted in working , $\div 20$ and then given to 2 sig fig, e.g. 150	
ecf for everything correct with an incorrect measurement shown in working (remember 5 is rounded down OR up) e.g. $45\,000 \div 20 = 2\,300$ OR e.g. $110 \div 0.02 = 5500$	1500 1.5×10^3
correct answer with incorrect unit	
29 000 to 30 000 (μm) 2.9×10^4 to 3.0×10^4	

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2(b)	An explanation that makes reference to three of the following points: • {generative nucleus / it} (divides to) form two (male) {nuclei / gametes} (1) • double fertilisation occurs (1) • one (male) {nucleus / gamete} fuses with {female / egg cell} nucleus to result in {an embryo / a zygote} (1) • one (male) {nucleus / gamete} fuses with (two) polar nuclei to result in endosperm formation (1)	ignore sperm cells ignore by meiosis reject generative nucleus reject generative nucleus	(3)

Question Number	Answer	Additional guidance	Mark
2(c)(i)	An answer that makes reference to the following point: • (loci of) {the genes / flower colour and pollen grain size (genes)} being on same {chromosome / chromatid} (1)	accept alleles for genes accept these genes inherited {as single unit / together} ignore they are at the same locus	(1)

Question Number	Answer	Mark
2(c)(ii)	The only correct answer is A Chi squared <i>B is not correct because correlation coefficient does not analyse observed phenotype frequencies and expected phenotype frequencies.</i> <i>C is not correct because standard deviation does not analyse observed phenotype frequencies and expected phenotype frequencies.</i> <i>D is not correct because Student's t-test does not analyse observed phenotype frequencies and expected phenotype frequencies.</i>	(1)